

YANHAN RUAN

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👤 PROFESSIONAL SUMMARY

Full-stack software engineer specializing in React, TypeScript, and Go, with a strong focus on e-commerce platforms and system architecture. Passionate about tackling complex PostgreSQL database designs, concurrent operations, and cross-platform UI challenges. Highly motivated to thrive in an English-first engineering environment, with a strong interest in adopting Elixir and functional programming to elegantly solve scalable SaaS business problems.

👥 EXPERIENCE & PROJECTS

e-idol (E-commerce & Gaming Marketplace), Tokyo, Japan Oct.2025 – Now

React | Tailwind CSS | Go (Gin) | PostgreSQL | WebSockets | AWS Personal Project

- **Architected a full-stack e-commerce marketplace** catering to specific subcultures, managing complete order lifecycles, user authentication (JWT/RBAC), and secure client-host interactions.
- **Engineered a concurrent booking and inventory system** utilizing database-level transactional locks (FOR UPDATE) in PostgreSQL, successfully eliminating race conditions and double-booking during peak traffic.
- **Built real-time communication** via native Go WebSockets (Hub/Client model) for instant messaging, live order state updates, and seamless customer interactions.
- **Deployed scalable infrastructure** using Docker and AWS (EC2, RDS) to ensure high availability, leveraging ongoing AWS Solutions Architect (SAA) principles to optimize cloud resource management.
- **Optimized frontend rendering** by developing a custom HTML5 Canvas engine with `requestAnimationFrame`, replacing heavy third-party animation libraries.
- Implemented a context-based **i18n** system enabling seamless multilingual localization for international users.

shadcn/ui | *Component Contributor*

Feb.2026

TypeScript | React | Tailwind CSS | Playwright Open Source

- **Resolved a critical Safari rendering collapse** in the Calendar component by refactoring the hybrid layout architecture from Flexbox to CSS Grid, ensuring pixel-perfect cross-browser consistency.
- **Engineered a cross-platform debugging workflow** utilizing Playwright's WebKit engine on Windows to isolate macOS-specific CSS anomalies, culminating in a successful Pull Request to the repository.

Carbon Neutralization Big Data Platform, Shanghai, China

Jun.2024 – Sept.2025

TypeScript | React | Express | Three.js | Resium FullStack Developer

- **Optimized algorithm performance** by implementing a dual-pointer approach, reducing time complexity from $O(n^2)$ to $O(n)$ and accelerating remote sensing image preprocessing by 150%.
- **Enhanced rendering efficiency** by encapsulating reusable utility hooks, throttling Cesium re-render events, and optimizing entity loading within specific viewport heights.
- **Secured application components** by developing robust Higher Order Components (HOC) for streamlined login validation and authentication routing.

🎓 EDUCATION

Wuhan University (WHU), Wuhan, China

Sept.2022 – Dec.2024

Master of Science degree in Digital Water Systems and Intelligent Solutions

Northwest A&F University (NWAUFU), Shaanxi, China

Sept.2018 – Jul.2022

Bachelor of Science degree in Water Resources and Hydropower Engineering

SKILLS & AWARDS

- **Languages & Frameworks:** TypeScript, ECMAScript 6+, React, Next.js, Go, Gin, Node.js, Python. (Eager to learn and transition to Elixir).
- **Databases & Cloud:** PostgreSQL, AWS, Docker, CI/CD.
- **Specialized Technologies:** Tailwind CSS, shadcn/ui, WebSockets, HTML5 Canvas, WebGIS.
- 2022 ZJU PAT - Professional Ability Test (95 / 100 score)
- 2022 ACM-ICPC excellence-award
- 2020 MCM S-Award
- 2020 CMC second-prize

LANGUAGES

- English: TOEIC 805; Professional proficiency for workplace communication and technical documentation.
- Japanese: JLPT N2 certified; conversational proficiency for daily communication.